Engineering Mechanics 1 · Unit_02 · UNIT 2

DRAFT — UE26ME142A · Unit_02 · UNIT 2 · UE26ME142A Unit 2 Class 23 AV Summary

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · UE26ME142A Unit 2 Class 23 AV Summary
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

UE26ME142A Unit 2 Class 23 AV Summary

1 UE26ME142A Unit 2 Class 23 AV Summary

Study focus: UE26ME142A_Unit 2_Class 23_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000222 · EV0000223

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Sv Vength, Civil Engineering, Department of Science and Humanities, PS University.
- This is class number thirty-two.
- We are in the process of solving some problems.
- In chapter four, unit two, that is on trusses.
- In the last class we had solved one problem.
- Today we will be continuing and solving some more problems.
- This is I've taken problem number four bar eight.
- The statement of the problem goes like this.
- Determine the force in each member of the loaded truss.
- You have load at A, which is twelve kilonewtons, and load at E which is eighteen kilonewtons.
- B D and B C as well as C D.
- So to find out the forces in this member, first thing that we have to do is determine the angle made by the member AB with the horizontal that is angle BAE.
- which we will get it using this triangle ABE.
- Sorry about this should be tan theta 2.
- Because we are assuming it to be under tension, internal forces in case of tension will be acting towards each other.
- Sine of angle theta 1 should be equal to 12 kilonewtons, that is what is mentioned here.
- Next we take some of all the forces along the x direction.
- So we get Fe to be equal to minus 19.2 kilovattan, which means that it is under compression.

- However we will keep the sign as it is as minus and take that force to.
- B in the member A E to be in tension only.
- Next coming to joint E.
- If I take joint E, the force in the member E B will be equal to 18 kilonewtons, that's what is mentioned here, which will be under tension itself.
- Then you take summation of all the forces along X.
- We have got AE as negative, so the force in the member E D will also be negative, which indicates that it is under compression, which will be of the magnitude nineteen point two itself here also.
- Next we will take the joint B.
- We have determined these two joints already, we know the these two members, sorry, not the joint.
- The forces in these two members is known, so we can find out the force in these two members.

Example or application from the lecture

- Consider all the members to be under tension, that means we are taking the forces in each of the member moving away from the joint.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000222 · EV0000223 · EV0000224 · +1 source units in pack.json

Core explanation · 05:00–09:59

as well as along horizontal.

Key details

- Sigma F y equals 0.
- Whereas this would be negative, so minus FP.
- B D also will be negative minus F D sine twenty two point six one which is this angle.
- Then we have the force in the member BA, which will also be negative, which is FAB sin 32.
- That is what is indicated here and also the component of the force B a along x.
- BA along x will be negative, so that is what is indicated here.
- Whereas this two is going to be positive because of these two forces.
- F B D in this equation, we get the force in the member B C which is positive sixty five point nine four kilonewton.
- So that means.
- The force in B C will be tensile force of magnitude 65.94.
- Substituting this value back in this equation, we will get the force in the member Bd as minus 50.05 kilonewton.
- Only one member that is left is this member that is C D.
- And we can we know the force in this member, this member.
- some of the forces that we have got it as positive, some of the forces we have got it as negative.
- You can indicate the force in the member if it is tensile, it is acting towards the joint.

- It is acting away from the joint.
- Or there is another way of indicating the forces in the members with respect to a table.
- That table is going to be looking something like this, where you will have serial number, member.
- That is the name of the member and what is the force magnitude that is acting in that member and what is the type of force that is acting, whether it is compressive or tension.
- A B A E B A E Oh there is a mis equal to.
- AE is equal to E D, I think this should be E D.
- Then we have B C and B D and C D, and corresponding values of the forces that we have got it to while we have analysed it is what is mentioned along with the sign.
- If it is negative, we will write it as compression.
- Or why should we indicate whether it is tension and compression?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000226 · EV0000227 · EV0000228 · +1 source units in pack.json

Core explanation · 10:00–14:59

That determination is different for a compression member and it is different for a tension member.

Key details

- That's why we should know whether the member is under compression or tension.
- However, in the first year syllabus, which is common for all engineering branches, we just have to determine what is the magnitude of the force and what is the type of the force.
- We will be stopping at this point.
- Further design of the members will be taken by the civil engineering in their IR submission.
- This problem is problem number four bar ten.
- Again, we have taken these problems from the prescribed textbook that is Engineer Mechanics Statics by Miriam and Craig and adopted by Colden, 8th edition only, or eighth volume.
- Or eighth edition only is what we have adopted.
- The problem statement goes something like this.
- Determine the force in each member of the loaded truss.
- Make use of the symmetry of the truss and of the loading.
- What is that hint?
- It says that make use of the symmetry of the truss and the loading.
- What does this mean?
- No, we say that a trust is symmetrical.
- Three conditions are satisfied.
- What are these three conditions?
- The three conditions is if you identify a truss having an symmetric axis, then the truss should be symmetric in.
- Geometry to the left of the section as well as to the right of the section.
- As to the light, right?
- That is the first condition.

- The second condition is the loading also should be symmetrical with respect to the symmetric axis.
- That means we have a loading at turn.
- As 30 kilometer, which is to the left of the symmetric axis Cg, which is at a distance of 5 meters from G.
- However, we have a loading at G itself and that is along the symmetric axis itself, so it is symmetrical.
- Okay, so two conditions one is symmetry with respect to geometry, the second condition is symmetry with respect to the loading.
- The last condition is it should be symmetrical with respect to the reactions also.
- Note, I'm telling it should be symmetrical with respect to reactions also.
- What is this?
- We get the reaction because of the supports.
- One vertical and one horizontal is what we take, but E we take it as one reaction which will be perpendicular to the line of the roller, which is in this case is line of roller is horizontal, so your reaction will be acting vertical.
- If you absorb the entire truss, we don't have any horizontal load acting external.
- Therefore, horizontal reaction at A is all zero.
- Since the horizontal reaction at A is zero, we will have only vertical reaction at A as well as vertical reaction at E.
- And this is symmetrical with respect to the symmetric axis.
- That means from G the reaction or the load at A is at a distance of ten meters.
- The advantage of a symmetrical truss is that.
- Whatever forces that we get to the left of the symmetric axis in the members, in all the members, the same forces will be acting to the right side also.
- That is the advantage.

Example or application from the lecture

- Let's go to the next problem.
- We have to find out the forces in all the members, but he has given you an hint.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000230 · EV0000231 · EV0000232 · +1 source units in pack.json

Core explanation · 15:00–19:59

If I could find out the forces in this six members, one, two, three, four, five, seven members.

Key details

- One, two, three, four, five, six and seven members, the forces in the member GF will be equal to GH.
- CF will be equal to CH, C D will be equal to C B, then your DE will be equal to PA, then your F E will be equal to H G.
- The next thing.
- How do we solve this?

- That's explanation for a symmetrical process.
- As for the method of joints, we have to identify a joint which has got only two unknowns.
- But if you observe here, we cannot find out any.
- Joint which has got only two unknowns, there are more than two unodes.
- But if I could somehow determine the reaction at A and reaction at E, then I can start with joint A or joint E.
- The vertical reactions will also be symmetrical.
- Therefore, I can easily say that reaction at A is equal to reaction at E, which will be equal to total vertical forces that is acting divided by 2.
- What is the total vertical force that is acting?
- That's why we get the reaction at A along the vertical and the reaction at E to be equal to 60 kilowattons.
- And as I have already explained, if you take sigma f y equals 0, you get the reaction at.
- AE at that is the horizontal reaction at A will be equal to zero.
- Next thing is we have to determine what is the angle made by the member AB with respect to AH or the horizontal, that is the angle PAH, which we can take this triangle and find out this angle, which I if I take it as theta, then it will be done inverse of 4 by 3 which is indicated here and that angle will be 38 point so.
- Compressive and the force in the member AH as tensile, then that's why you can see here force in the member AH is moving away from the joint, force in the member AB is moving towards the joint.
- Sigma F equals 0, sigma f x equals 0.
- If I take sigma f x equals 0, I will get the force in the member AH.
- You can look at the value of this in the PPT that has been attached in PSO Academy.
- So continuing this, next we will take joint B.
- In joint B again, we have assumed the force in the member.
- A B as compressive, so here also we will be assuming the same thing to be moving toward the joint.
- Whereas at B the force is moving away from the joint, we are assuming it to be under tension, this also the force in B H also as tensile force.
- We had got the value of the force in the member AB as 96 kilonewton, which is compressive, and that is positive, we have got.
- And then now we will apply the apply the equations of equilibrium after resolving this 96 kilometer force along vertical and horizontal.
- Next you take joint H in joint H.
- We have got the value of P C which is seventy-five.
- We have got it as negative.
- Then 60 kilon is the force acting upwards, 30 kilonewton is acting downwards.
- How did we get this 60?
- We have got it as BHS sixty tensile, so that is as it is.
- Then these two are to be determined.
- I get the value of CH and GH and this is what the values that we have got.
- Then CG is assumed to be tensile, we get it as positive and that is why it is tensile.
- Next you can take joint at G.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000234 · EV0000235 · EV0000236 · +1 source units in pack.json

Core explanation · 20:00–24:59

Joint G we have four members.

Key details

- And it will be under tension.
- We will continue in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000238 · EV0000239 · EV0000240 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 2_Class 23_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/023_UE26ME142A_Unit_2_Class_23_AV_Summary__d2db844898691c1a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 23_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/023_UE26ME142A_Unit_2_Class_23_AV_Summary__d2db844898691c1a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 23_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/023_UE26ME142A_Unit_2_Class_23_AV_Summary__d2db844898691c1a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 23_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/023_UE26ME142A_Unit_2_Class_23_AV_Summary__d2db844898691c1a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md have no readable `.slides.md` text; their original page images are included.
2. 4 rendered pages in 015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md have no readable `.slides.md` text; their original page images are included.
4. 4 rendered pages in 017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md have no readable `.slides.md` text; their original page images are included.
7. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md have no readable `.slides.md` text; their original page images are included.
9. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md have no readable `.slides.md` text; their original page images are included.
10. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md have no readable `.slides.md` text; their original page images are included.

11. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md have no readable `.slides.md` text; their original page images are included.
12. 3 rendered pages in 021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md have no readable `.slides.md` text; their original page images are included.
14. 3 rendered pages in 023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md have no readable `.slides.md` text; their original page images are included.
16. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
17. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
18. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
19. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
20. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00054	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...
SRC00055	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00056	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...
SRC00057	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...

C Revision and retrieval practice